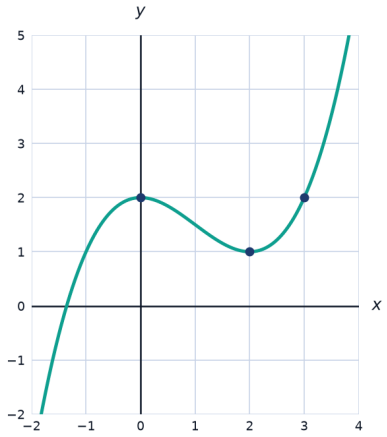
Radicals & Rational Exponents

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Inverse relationship between exponentials and logarithms

For the model $f(x) = 5x + 4$ in inverse relationship between exponentials and logarithms, evaluate $f(5)$.

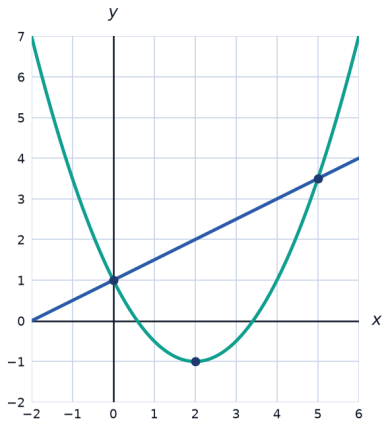


2 [Foundational] • Target: Inverse relationship between exponentials and logarithms

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

3 [Foundational] • Target: Logarithmic functions

For the model $f(x) = 6x + 5$ in logarithmic functions, evaluate $f(2)$.



4 [Foundational] • Target: Logarithmic functions

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

5 [Foundational] • Target: Logarithmic functions

A data set recorded for logarithmic functions consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean? Show your complete algebraic work.

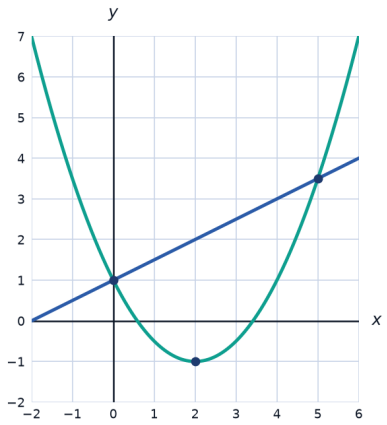
© ClipSAT — clipsat.org | All Rights Reserved

6 [Foundational] • Target: Logarithmic functions

For the model $f(x) = 6x + 5$ in logarithmic functions, evaluate $f(3)$.

7 [Foundational] • Target: Logarithmic functions

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Logarithmic functions

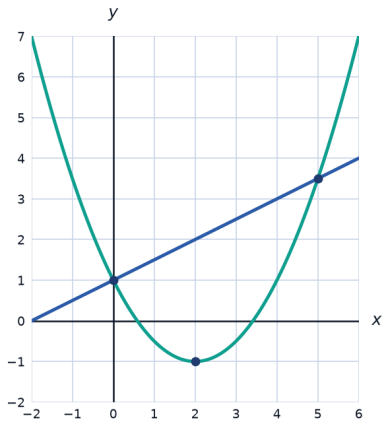
For the model $f(x) = 6x + 5$ in logarithmic functions, evaluate $f(4)$.

9 [Intermediate] • Target: Logarithmic functions

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

10 [Intermediate] • Target: Logarithmic functions

For the model $f(x) = 6x + 5$ in logarithmic functions, evaluate $f(5)$.



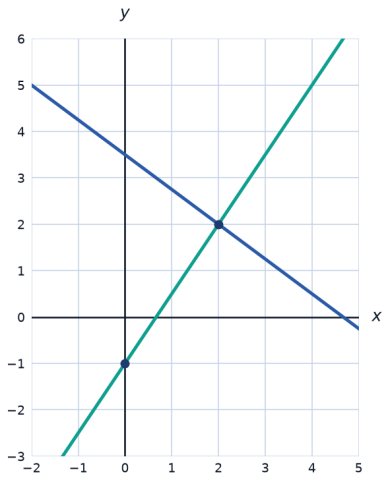
11 [Intermediate] • Target: Logarithmic functions

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

© ClipSAT — clipsat.org | All Rights Reserved

12 [Intermediate] • Target: Extension: Properties of logarithms

For the model $f(x) = 2x + 6$ in extension: properties of logarithms, evaluate $f(2)$.



13 [Intermediate] • Target: Extension: Properties of logarithms

A data set recorded for extension: properties of logarithms consists of the four values $\{10, 14, 18, 24\}$.

What is the arithmetic mean? Show your complete algebraic work.

14 [Intermediate] • Target: Extension: Properties of logarithms

For the model $f(x) = 2x + 6$ in extension: properties of logarithms, evaluate $f(3)$.

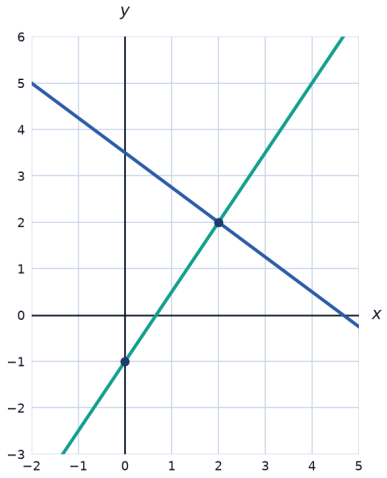
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Extension: Properties of logarithms • Cross-Lesson: 10.01 Matrices & Systems

For the model $f(x) = 2x + 6$ in extension: properties of logarithms, evaluate $f(4)$.

16 [Challenging] • Target: Extension: Properties of logarithms • Cross-Lesson: 11.01 Probability & Statistics

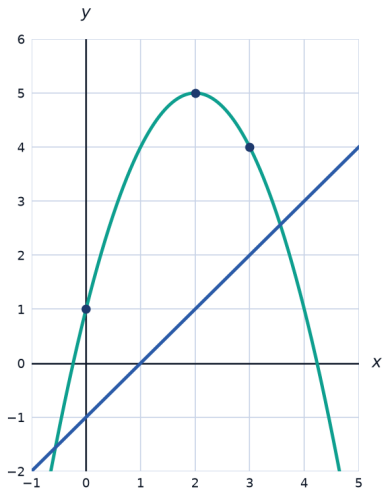
For the model $f(x) = 2x + 6$ in extension: properties of logarithms, evaluate $f(5)$.



© ClipSAT — clipsat.org | All Rights Reserved

17 [Challenging] • Target: Exponential and logarithmic equations • Cross-Lesson: 2.01 Quadratics & Complex Numbers

For the model $f(x) = 3x + 7$ in exponential and logarithmic equations, evaluate $f(2)$.



18 [Challenging] • Target: Exponential and logarithmic equations • Cross-Lesson: 3.01 Polynomial Functions

A data set recorded for exponential and logarithmic equations consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean? Show your complete algebraic work.

19 [Challenging] • Target: Exponential and logarithmic equations • Cross-Lesson: 4.01 Rational Expressions

For the model $f(x) = 3x + 7$ in exponential and logarithmic equations, evaluate $f(3)$.

20 [Challenging] • Target: Exponential and logarithmic equations • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

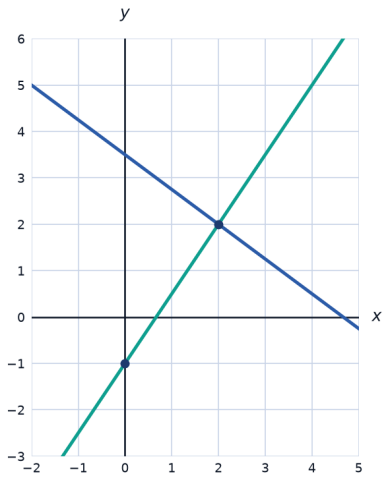
For the model $f(x) = 3x + 7$ in exponential and logarithmic equations, evaluate $f(4)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

© ClipSAT — clipsat.org | All Rights Reserved

21 [Foundational] • Target: Graphs and characteristics of exponential functions

For the model $f(x) = 3x + 9$ in graphs and characteristics of exponential functions, evaluate $f(2)$.



a. 15

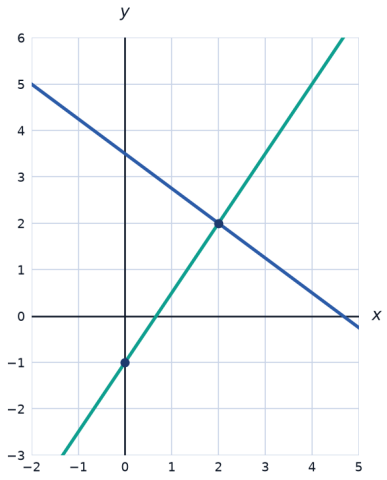
b. 18

c. 6

d. 19

22 [Foundational] • Target: Graphs and characteristics of exponential functions

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.



a. $x = 2$ and $x = 6$

b. $x = 2$ and $x = 5$

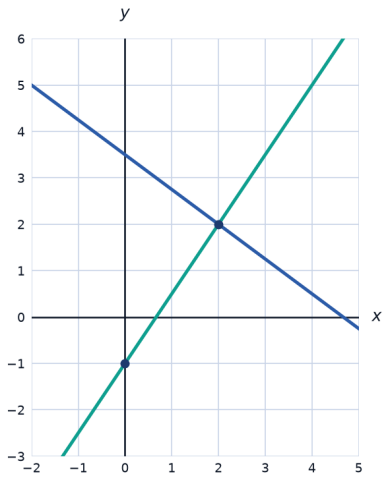
c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

© ClipSAT — clipsat.org | All Rights Reserved

23 [Foundational] • Target: Graphs and characteristics of exponential functions

For the model $f(x) = 3x + 9$ in graphs and characteristics of exponential functions, evaluate $f(5)$.



a. 24

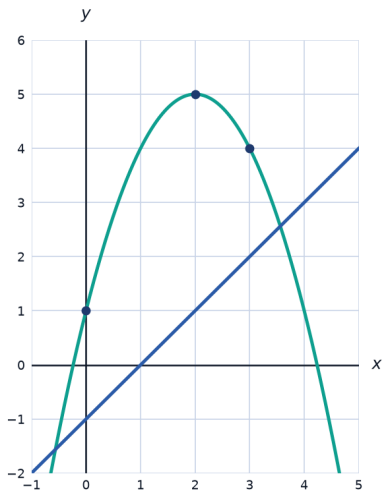
b. 27

c. 15

d. 28

24 [Foundational] • Target: Applications of exponential functions

For the model $f(x) = 4x + 3$ in applications of exponential functions, evaluate $f(2)$.



a. 11

b. 15

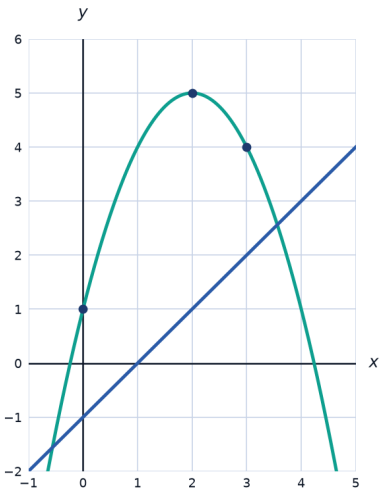
c. 8

d. 15

© ClipSAT — clipsat.org | All Rights Reserved

25 [Foundational] • Target: Applications of exponential functions

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.



a. $x = 2$ and $x = 7$

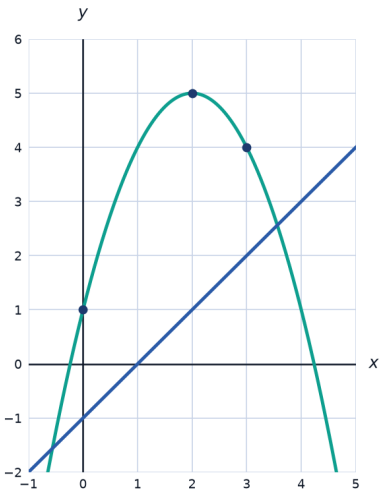
b. $x = 2$ and $x = 6$

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

26 [Foundational] • Target: Applications of exponential functions

For the model $f(x) = 4x + 3$ in applications of exponential functions, evaluate $f(5)$.



a. 23

b. 27

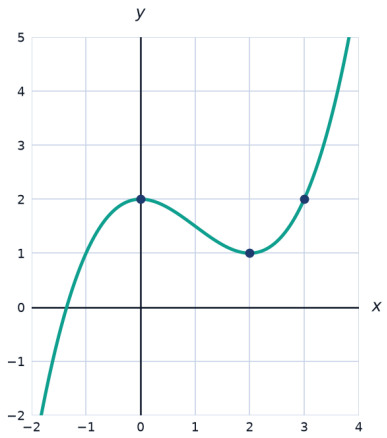
c. 20

d. 27

© ClipSAT — clipsat.org | All Rights Reserved

27 [Foundational] • Target: Inverse relationship between exponentials and logarithms

For the model $f(x) = 5x + 4$ in inverse relationship between exponentials and logarithms, evaluate $f(2)$.



a. 14

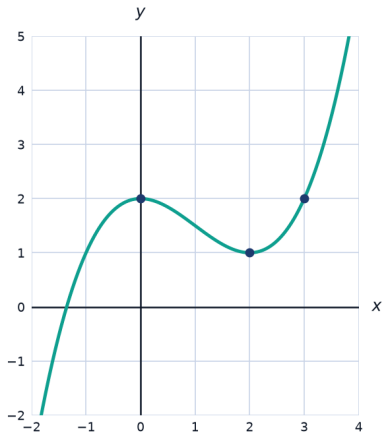
b. 19

c. 10

d. 18

28 [Foundational] • Target: Inverse relationship between exponentials and logarithms

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.



a. $x = 2$ and $x = 8$

b. $x = 2$ and $x = 7$

c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

© ClipSAT — clipsat.org | All Rights Reserved

29 [Intermediate] • Target: Graphs and characteristics of exponential functions

Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

a. $x = 1$ and $x = 5$

b. $x = 1$ and $x = 4$

c. $x = -1$ and $x = -4$

d. $x = 5$ and $x = 4$

30 [Intermediate] • Target: Graphs and characteristics of exponential functions

A data set recorded for graphs and characteristics of exponential functions consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5

d. 23.5

31 [Intermediate] • Target: Graphs and characteristics of exponential functions

For the model $f(x) = 3x + 9$ in graphs and characteristics of exponential functions, evaluate $f(3)$.

a. 18

b. 21

c. 9

d. 22

32 [Intermediate] • Target: Graphs and characteristics of exponential functions

For the model $f(x) = 3x + 9$ in graphs and characteristics of exponential functions, evaluate $f(4)$.

a. 21

b. 24

c. 12

d. 25

33 [Intermediate] • Target: Graphs and characteristics of exponential functions

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ and $x = 6$

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

© ClipSAT — clipsat.org | All Rights Reserved

34 [Intermediate] • Target: Graphs and characteristics of exponential functions

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ and $x = 7$

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

35 [Intermediate] • Target: Applications of exponential functions

A data set recorded for applications of exponential functions consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5

d. 24.5

36 [Intermediate] • Target: Applications of exponential functions

For the model $f(x) = 4x + 3$ in applications of exponential functions, evaluate $f(3)$.

a. 15

b. 19

c. 12

d. 19

37 [Intermediate] • Target: Applications of exponential functions

For the model $f(x) = 4x + 3$ in applications of exponential functions, evaluate $f(4)$.

a. 19

b. 23

c. 16

d. 23

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Applications of exponential functions • Cross-Lesson: 3.01 Polynomial Functions

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ and $x = 7$

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

© ClipSAT — clipsat.org | All Rights Reserved

39 [Challenging] • Target: Applications of exponential functions • Cross-Lesson: 4.01 Rational Expressions

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

a. $x = 4$ and $x = 9$

b. $x = 4$ and $x = 8$

c. $x = -4$ and $x = -8$

d. $x = 12$ and $x = 32$

40 [Challenging] • Target: Inverse relationship between exponentials and logarithms • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

a. $x = 1$ and $x = 7$

b. $x = 1$ and $x = 6$

c. $x = -1$ and $x = -6$

d. $x = 7$ and $x = 6$

41 [Challenging] • Target: Inverse relationship between exponentials and logarithms • Cross-Lesson: 7.01 Sequences & Series

For the model $f(x) = 5x + 4$ in inverse relationship between exponentials and logarithms, evaluate $f(3)$.

a. 19

b. 24

c. 15

d. 23

42 [Challenging] • Target: Inverse relationship between exponentials and logarithms • Cross-Lesson: 8.01 Conic Sections

For the model $f(x) = 5x + 4$ in inverse relationship between exponentials and logarithms, evaluate $f(4)$.

a. 24

b. 29

c. 20

d. 28

43 [Challenging] • Target: Inverse relationship between exponentials and logarithms • Cross-Lesson: 9.01 Trigonometric Functions

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ and $x = 8$

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

© ClipSAT — clipsat.org | All Rights Reserved

44 [Challenging] • Target: Logarithmic functions • Cross-Lesson: 1.01 Functions & Their Graphs

A data set recorded for logarithmic functions consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5

d. 26.5

45 [Challenging] • Target: Extension: Properties of logarithms • Cross-Lesson: 10.01 Matrices & Systems

A data set recorded for extension: properties of logarithms consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

d. 18.5

© ClipSAT — clipsat.org | All Rights Reserved